

IT462 — Exploratory Data Analysis — Lab 7

Employee Attrition Dataset: A Complete EDA Workflow

General Instructions

- Use Python with pandas, numpy, matplotlib, and seaborn (or plotly) for all tasks.
- Answer every question inside a single Jupyter Notebook, in the same order as given below.
- Each code cell must be followed by a short markdown cell containing your written observation or interpretation, unless the question states otherwise.
- All plots must have a title, labeled axes, and a legend where applicable.
- Do not delete or hide any intermediate output — your reasoning matters as much as your final answer.
- Clearly mention any assumptions you make while handling missing values or outliers.
- Rename your final notebook as **RollNumber_Name_Lab7.ipynb** before submission.

Dataset Description

You are provided with a synthetically generated Human Resources dataset named **employee_eda_dataset.csv**, representing 1000 employees of a mid-sized company. The dataset intentionally contains missing values and outliers so that you can practice the complete EDA workflow. The columns are described below.

Column	Description
EmployeeID	Unique identifier for each employee
Age	Age of the employee in years
Gender	Male or Female
Department	Department the employee works in
EducationLevel	Highest education qualification
YearsAtCompany	Tenure at the company in years
MonthlyIncome	Monthly salary in local currency
DistanceFromHome_km	Distance between home and office in kilometers
WorkLifeBalanceScore	Self-reported score from 1 (poor) to 5 (excellent)
JobSatisfaction	Self-reported score from 1 (low) to 5 (high)
NumProjects	Number of projects currently assigned
OvertimeHoursPerMonth	Average overtime hours worked per month
TrainingHoursLastYear	Training hours completed in the last year
PerformanceRating	Annual performance rating from 1 to 5
Attrition	Whether the employee left the company (Yes/No)

Question 1 | Data Loading and Initial Exploration

- (a) Load the dataset `employee_eda_dataset.csv` into a pandas DataFrame and display the first 10 and last 5 rows.
- (b) Report the shape of the dataset (number of rows and columns). State how many numerical and how many categorical columns are present.
- (c) Use the `info()` function to display the data types and non-null counts of every column. Comment on which columns appear to have missing data based on this output.
- (d) Use the `describe()` function to generate summary statistics for all numerical columns, and `describe(include='object')` for the categorical columns. Based on the mean, standard deviation, minimum, and maximum values, identify at least three columns that look suspicious or potentially contain outliers, and justify your answer using the specific numbers you observe.
- (e) Check for and report the number of duplicate rows in the dataset, if any. If duplicates exist, remove them and confirm the new shape of the dataset.
- (f) Print the `value_counts` for each categorical column (Gender, Department, EducationLevel, Attrition) and comment on whether any category looks unbalanced.

Question 2 | Missing Value Handling

- (a) Compute the total number of missing values in each column, both as a raw count and as a percentage of total rows. Present this as a table sorted from highest to lowest percentage of missingness.
- (b) Create a visualization to display the pattern of missing values across the dataset (for example, a heatmap using seaborn, or a bar chart of missing value counts). Write two observations from this visualization.
- (c) For at least three columns with missing values, discuss whether the missingness is likely to be Missing Completely at Random (MCAR), Missing at Random (MAR), or Missing Not at Random (MNAR), with justification.
- (d) Handle the missing values using at least three different techniques across different columns and justify each choice. Your treatment must include: (i) deletion, applied where appropriate; (ii) simple imputation using mean or median for a numerical column; (iii) categorical imputation for EducationLevel; and (iv) one advanced technique such as group-wise imputation, KNN imputation, or regression-based imputation for at least one column.
- (e) After treatment, verify that there are no remaining missing values, and compare the mean and standard deviation of at least one numerical column before and after imputation to check whether the original distribution was distorted.

Note: You do not need to use the same imputation technique for every column. Different columns may call for different strategies depending on their distribution and relationship with other features.

Question 3 | Outlier Detection, Visualization and Treatment

- (a) For each numerical column, create box plots to visually identify the presence of outliers, arranged in a grid layout for readability. Also create histograms and/or KDE plots for at least four numerical columns and comment on their skewness.

- (b) Detect outliers using the Interquartile Range (IQR) method for at least four numerical columns. For each column, report the lower bound, upper bound, and the number of records identified as outliers.
- (c) Detect outliers using the Z-score method for the same columns used in part (b), using a threshold of your choice (commonly 3). Compare the number of outliers detected by each method and explain any differences.
- (d) Create a scatter plot between two numerical columns of your choice (for example, MonthlyIncome and YearsAtCompany) and visually mark or highlight the points flagged as outliers.
- (e) Treat the outliers using at least three different strategies across different columns and justify each choice. Your treatment must include: (i) removal, for a column where the outliers represent invalid or impossible data; (ii) capping or winsorization, for a column where outliers may represent genuine but extreme values; and (iii) a mathematical transformation (such as log or square root) applied to reduce skewness in one column, shown before and after with histograms.
- (f) After treatment, regenerate the box plots for the treated columns and summarize how the treatment changed the spread and shape of each distribution.

Note: When deciding whether an outlier should be removed, capped, or kept, always consider whether the value is a genuine business possibility or a data entry error. Explain this reasoning explicitly in your markdown cells.

Question 4 | Correlation Analysis

- (a) Compute the correlation matrix for all numerical columns using the Pearson correlation coefficient and display it as an annotated heatmap.
- (b) Identify the pair of features with the strongest positive correlation and the pair with the strongest negative correlation. Report the coefficients and give a plausible business explanation for each relationship. Also identify at least two pairs of features with very weak or negligible correlation and explain what this indicates.
- (c) Create a pair plot (seaborn pairplot) for at least five numerical columns, colored by the Attrition column, and write down at least two patterns or separations you observe between employees who left and those who stayed.
- (d) Compute the correlation of each numerical feature with the target variable Attrition (encode Attrition as 0/1 first). Rank the features from most to least correlated and discuss which features might be useful predictors of attrition.
- (e) Compare the Pearson correlation matrix with the Spearman correlation matrix for the same columns, and explain any notable differences and why Spearman correlation might behave differently in the presence of outliers or non-linear relationships.

Question 5 | Feature Engineering

- (a) Create a new feature called **AgeGroup** by binning the Age column into meaningful categories (for example, '20-30', '31-40', '41-50', '51+') and display the count of employees in each group.
- (b) Create a new feature called **IncomePerYearOfExperience** by dividing MonthlyIncome by (YearsAtCompany + 1), and explain what business insight this new feature might capture that the original columns do not.
- (c) Encode the categorical variables using an appropriate technique for each column (for example, One-Hot Encoding for nominal variables such as Department, and Label or Ordinal Encoding for ordinal

variables such as EducationLevel), and justify your choice for at least two columns.

(d) Create a new binary feature called **IsOverworked** that flags employees whose OvertimeHoursPerMonth is above a threshold you define, stating how you chose the threshold (for example using a percentile of the distribution).

(e) Scale the numerical features using both Min-Max Scaling and Standardization (Z-score scaling). Display the summary statistics of one column before and after each method, and explain when you would prefer one scaling method over the other.

(f) Using your engineered features along with the original numerical features, create one final visualization that best summarizes a relationship or insight related to employee attrition, and justify your choice.

(g) Summarize your entire EDA process in a short written report (250 to 400 words) covering the key data quality issues found, how missing values and outliers were handled, the most important correlations discovered, and the new features you believe would be most useful for a future predictive model of attrition.

End of Lab 7 — Please ensure your notebook runs top to bottom without errors before submission.